

Abdominal FAST

Review Video: <https://coreultrasound.com/efast>

Indications:

- Blunt/penetrating abdominal injury
- Blunt/penetrating thoracic injury

Knobology and Preparation:

- Depth: **20cm** or machine max
- Probe: curvilinear
- Gain: midrange
- Systems presets: abdominal

Upper Quadrants:

- **External Landmark:** mid to posterior axillary line at the level of the xiphoid
- **Internal Landmark:** kidney
- **AOI:** solid organ/renal interface (between capsule of liver/spleen and Gerota's fascia), including the tip of the solid organ, subdiaphragm on the L. side

Upper quadrants

- Place probe at **mid to post axillary line at the level of the xiphoid process** in the **longitudinal** plane, indicator pointing **cephalad**
- Slide the probe posteriorly/anteriorly to locate the kidney and ensure the best possible view of the interface.
- Slide cephalad/caudad in a longitudinal plane to explore the interface, again keeping the probe parallel to the floor/stretchers.
 - o You are doing this to adequately explore the extent/size of the interface as well as to find the best window/s to interrogate the interface.
- Center and magnify the image
- **RUQ:**
 - o **Sweep through the hepatorenal interface**
 - o Identify **the caudal tip of the liver and interrogate for free fluid**
 - ***This is the most sensitive place to identify free fluid on the fast exam*
- **LUQ:**
 - o Sweep through the entire **splenorenal interface**
 - o Slide cephalad to visualize the **subdiaphragmatic space**. Ensure that you visualize the diaphragm from 6 to 9 o'clock and sweep through
 - The subdiaphragmatic space is the most likely place to find free fluid in the LUQ

Troubleshooting:

- **Ribs shadows:**
 - o Move in the longitudinal plane → can create **overlapping** images
 - o Probe rotation – can rotate the **probe indicator slightly down** towards the bed to try to get the probe between ribs
- Patient respirations
- **PALMS:**

- **PA:** move **p**osterior/**a**nterior to find interspace
- **L:** slide **L**ongitudinally to center image
- **M:** **M**agnify/optimize image
- **S:** **S**weep interface
- **LUQ Diaphragm**
 - 1. Find the best view of the interface/kidney
 - 2. Slide cephalad in longitudinal plain
 - 3. Probe rotation (deals with rib shadows)
 - 4. Slide anterior and sweep posteriorly (or vice versa)
 - 5. Slide caudad and heel through the spleen

Pitfalls:

- Image interpretation:
 - Attempting to interpret part of the interface that has a rib shadow persistently obscuring it.
 - Mistaking perinephric fat, blood vessel, gallbladder, stomach as free fluid
 - Mistaking other abdominal fluids for blood (ascites, urine, dialysate, intraluminal bowel fluid)
 - Mistaking the stomach wall for the diaphragm in the LUQ
- Clinical Integration
 - Assuming the absence of free fluid in an acute trauma patient rules out solid organ, hollow viscus or vascular injury.
 - Only can rule out significant hemoperitoneum (>250ml of fluid) in one specific moment in time
 - Declaring negative scan in patients who have had multiple prior surgeries
 - Relying on a negative scan in a trauma patient whose presentation is delayed
 - Not repeating the scan in high-risk patients
 - Assuming the free fluid is always blood

Tips and Tricks:

- **The kidney does not define the location of the interface, this is defined by the solid organ. The kidney also does not define the extent of the interface, also defined by solid organ (including the caudal tip)**
- Place patient in **Trendelenberg** - delivers fluid to the RUQ
- Serial scans - "when in doubt, repeat your scan"

Pelvic

Knobology:

- Curvilinear
- Depth: usually **15cm** in adults
- Gain: Mid Range
- Presets: Abdominal/FAST
- Transverse and longitudinal.

AOI: Rectovesical pouch in men and rectouterine pouch and vesicular uterine pouch in women.

External Landmark: symphysis pubis (midline, just cephalad)

Internal Landmark: bladder

Steps:

1. Place the probe in the longitudinal plane immediately cephalad to the **symphysis pubis**
2. Identify the internal landmark (the bladder)
 - Note that this can be difficult if the patient has recently emptied their bladder
3. Find the area of interest by heeling through the bladder in a caudad/cephalad direction
4. Sweep through the area of interest
 - In female, you are looking for the uterus to disappear
 - In males, you are sweeping through the bladder
5. Rotate the probe to sweep in the transverse plane.
 - Start at the level of the vaginal stripe in women
 - Start from the seminal vesicles in men

Troubleshooting:

- Ensure probe is placed right against symphysis pubis
- Press and hold to displace bowel gas
- Slide the probe off centre and heel/toe back (in transverse view) or sweep back (in longitudinal view)
- Fill the bladder (i.e. IV fluids / get patient to drink if not NPO)

Pitfalls

- Image interpretation
 - Misidentifying the seminal vesicles as free fluid (will look like a bow-tie)
 - Misidentifying the prostates as free fluid
- Clinical integration
 - Assuming that all pelvic free fluid is acute blood.
 - Assuming that fluid in the pelvis is only physiologic

Tips and Tricks:

- Place patient in **reverse Trendelenburg** to make pelvic FF more visible
- Turn down gain to prevent washout from bladder
- Evaluate each side of the bladder in turn to prevent missing fluid on the sides

